THE PARAMETER SPACE OF APOLLONIAN CIRCLE PACKINGS THROUGH MÖBIUS TRANSFORMATIONS

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Abstract. An Apollonian circle packing is a fractal created by starting with three pairwise tangent circles and iteratively filling in circle tangent to all three circles. Using positive semidefinite binary quadratic forms one can build the parameter space of Apollonian circle packings over $\mathbb{P}^1(\mathbb{C})$. We show that this method of parameter space is equivalent to inversion through a point and extend the group action on $\mathbb{P}^1(\mathbb{C})$ to cover all equivalent packings.

1. Introduction

Placing four mutually tangent circles in the complex plane with disjoint interior creates a *Descartes quadruple*. When you pick three of these circles, we get one circle tangent to all three circles that is not already there; add it to the packing. Iterating this process gives an *Apollonian circle packing* (ACP). If the initial four circles have integral curvatures (reciprical of the radius) then every circle in the packing has integral curvature. Graham, Lagarias, Mallows, Wilks, and Yan [GLM+03] explored the fundamental properties of these packings and sparked many more papers.

Kocik [Koc20] and Holly [Hol21] use $[0,1]^2$ as a parameter space for Apollonian circle packings and show that the depths of circles in these packings creates an interesting fractal. Rickards [Ric24] builds upon their work using binary quadratic forms to build a parameter space in $\hat{\mathbb{C}}$, which is the strip packing $(0,0,2,2)$. These show that the location of a point in the parameter space determines the type of packing it corresponds to, i.e. a bounded, strip, half-plane, or full-plane packing. Using the maps in [Ric24], one can compute a binary quadratic form from a point in the parameter space and use this form to find an associated Descartes quadruple that can generate an Apollonain circle packing. Instead of computing binary quadratic forms, we show that the same correspondence can be realized with an easier map: inverting the strip packing through the point. We conclude by extending the corresponding group action to cover the full Apollonian group in $\mathbb{P}^1(\mathbb{C})$ and binary quadratic forms.

2. Building the Parameter Space

We will use Rickards maps [Ric24] to build the parameter spcae of Apollonian circle packings. In order to do this we need to define the relationships between ACPs and binary quadratic forms and $\mathbb{P}^1(\mathbb{C})$. A binary quadratic form (BQF) is a function of the form $Q(x,y) = Ax^2+Bxy+Cy^2$ for $A,B,C \in \mathbb{R}$, not all zero. BQFs can be written as $Q = [A,B,C]$ and has discriminant $\Delta = B^2 - 4AC$. Rickards [Ric24] and Graham et al. [GLM+03] explain in more detail the connection between binary quadratic forms and ACPs. For now, we now there is a bijection between

BQF of the form $[n, A, B, C]$, called a BQF quadruple, where the discriminant is $-4n^2$ and Desacrtes quadruples where the first entry is fixed.

Theorem 2.1 ([Ric24])**.** *Let n be a fixed real number. Then the n-quadruples (n, a, b, c) biject with BQF quadruples $[n, A, B, C]$ via the correspondence*

$$\theta([n, A, B, C]) := (n, A - n, C - n, A + C - B - n).$$

$$\phi(n, a, b, c) := [n, n + a, n + a + b - c, n + b]$$

Furthermore, primitive integral Descartes quadruples biject with primitive integral BQF quadruples.

This means every Apollonian circle packing with a fixed circle can be associated with an equivalence class of BQF quadruples. So, if BQF quadruples are equivalent, then their associated packings are the same.

We can use BQF quadruples to map to $\mathbb{P}^1(\mathbb{C})$ by taking a root of the quadratic form.

Definition 2.2. Let $Q = [n, A, B, C]$ be a BQF quadruple. The function $AX^2 + BX + C$ has two roots in $\mathbb{P}^1(\mathbb{C})$ so define the map

$$\rho : Q \to \rho(Q)$$

$$[n, A, B, C] \to p_Q := \begin{cases} \frac{-B+2ni}{2A} & \text{if } A \neq 0 \\ \infty & \text{if } A = 0 \end{cases}$$

Note that if $n > 0$ then the point is in the upper half plane, if $n < 0$ the it is the lower half plane root, and if $n = 0$ there is a unique root in $\mathbb{P}^1(\mathbb{C})$.

Lemma 2.3 ([Ric24])**.** *The map from BQF quadruples modulo $\mathbb{R}^+$ scalars to $\mathbb{P}^1(\mathbb{C})$ by $\rho : Q \to p_Q$ is a bijection*

We can compute these maps explicitly. Given a BQF quadruple modulo $\mathbb{R}^+$ we have

$$[n, A, B, C] \sim [n, 1, \frac{B}{A}, \frac{C}{A}]$$

or if $A = 0$

$$[n, A, B, C] = [0, 0, 0, C] \sim [0, 0, 0, 1]$$

So without loss of generality we can assume any BQF quadruple is of the form $[n, 1, B, C]$ or $[0, 0, 0, 1]$. Then we get

$$\rho(Q) = \begin{cases} \frac{-B+2ni}{2} = \frac{B}{2} + ni & \text{if } Q \sim [n, 1, B, C] \\ \infty & \text{if } Q \sim [0, 0, 0, 1] \end{cases}.$$

This gives

$$\rho^{-1}(z) = \begin{cases} [y, 1, -2x, x^2 + y^2] & \text{if } z = x + iy \\ [0, 0, 0, 1] & \text{if } z = \infty \end{cases}$$

Now that we have a bijection with n-quadruples and BQF-quadruples as well as a bijection with BQF-quadruples and $\mathbb{P}^1(\mathbb{C})$ we now consider their composition.

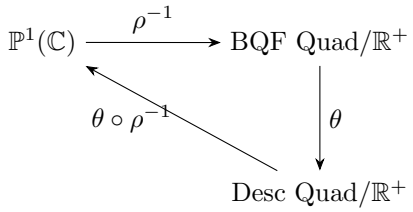


FIGURE 1. The bijective maps between $\mathbb{P}^1(\mathbb{C})$, BQF quadruples up to $\mathbb{R}^+$ scaling, and Descartes quadruples up to $\mathbb{R}^+$ scaling

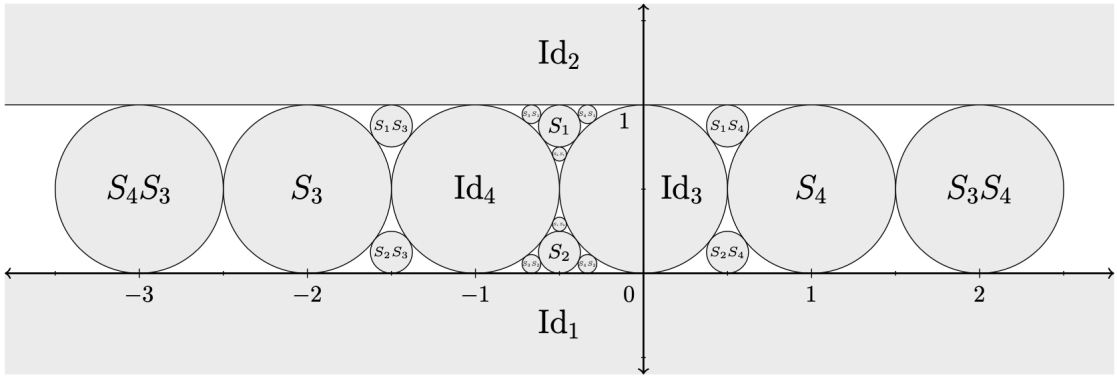


FIGURE 2. Depth circles labelled by depth element from [Ric24]

Lemma 2.4 (Rickards)**.** *There is a bijection between Descartes quadruples up to scaling by* $\mathbb{R}^+$ *and* $\mathbb{P}^1(\mathbb{C})$*. For Descartes quadruples up to scaling,* q*, the map to* $\mathbb{P}^1(\mathbb{C})$ *is given by*

$$\rho \circ \phi : q \to \rho(\phi(q))$$
$$(n,a,b,c) \sim \frac{1}{n+a}(n,a,b,c) \to \frac{n+a+b-c}{2(n+a)} + \frac{n}{n+a}i$$

Explicitly, we have the map

$$\theta \circ \rho^{-1} : \mathbb{P}^1(\mathbb{C}) \to \textit{ Descartes quadruples}/\mathbb{R}^+$$
$$(x+iy) \to (y, 1-y, x^2+y^2-y, 1+x^2+y^2+2x-y)$$

Figure 1 shows how these maps are interacting with each other. We will let $\psi = \theta \circ \rho^{-1}$ be the map from $\mathbb{P}^1(\mathbb{C})$ to Descartes quadrples up to scaling by $\mathbb{R}^+$. Now, consdiering the depth of a quadruple and graphing its depth circle (for more information see [Ric24]) this gives Figure 2.

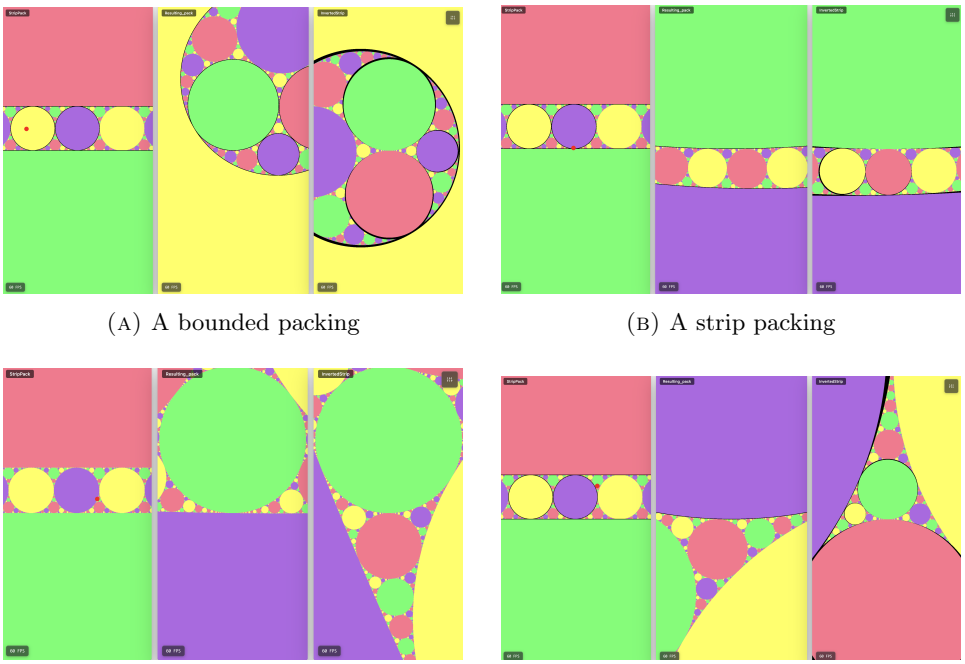
Remark 2.5. This fractal is also the analogue of the fractals in Kocik and Holly. Specifically, if q is a Descartes quadruple, then q generates a

- bounded packing if and only if $\psi^{-1}(q)$ is in the interior of a circle
- strip packing if and only if $\psi^{-1}(q)$ is a tangency point between two circles
- half-plane packing if and only if $\psi^{-1}(q)$ is on the boundary of a circle but not a tangent point

- full-plane packing if and only if $\psi^{-1}(q)$ is not contained in or on the boundary of a circle i.e. a limit point

3. Creating an Interactive Parameter Space

We can create a live rendering where a user can move their mouse in the parameter space and see the packing that the mouse point generates. This is made with shaders and can be found here https://summerhaag.github.io/demos/parameter-space/. While moving in the parameter space, this map starts to look very familiar. Figure 3 shows some screenshots from this visualization.



(A) A bounded packing

(B) A strip packing

(C) A half-plane packing

(D) A full-plane packing

Figure 3. Screenshots from the interactive parameter space, showing the packing generated by points in each of the four regions.

3.1. An equivalent map.

We will now show an equivalent map. Consider the relationship between mobius transformations and circles in $\hat{\mathbb{C}}$.

Proposition 3.1 ([Sta18]). *Let $z \to \frac{\alpha z+\beta}{\gamma z+\delta}$ be a mobius transformation. Then the image of $\hat{\mathbb{R}}$ under the transformation is a circle with curvature equal to $2\,\mathrm{Im}(\bar{\gamma}\delta) = i(\gamma\bar{\delta} - \bar{\gamma}\delta)$, and curvature-center equal to $i(\alpha\bar{\delta} - \bar{\gamma}\beta)$ and co-curvature is $2\,\mathrm{Im}(\alpha\bar{\beta} - \bar{\alpha}\beta)$.*

Using this, define the four circles in the strip packing $(0, 0, 2, 2)$ as the following:

Definition 3.2.

$$C_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \hat{\mathbb{R}}$$
$$C_2 = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix} \hat{\mathbb{R}}$$
$$C_3 = \begin{pmatrix} 1 & 0 \\ -i & 1 \end{pmatrix} \hat{\mathbb{R}}$$
$$C_4 = \begin{pmatrix} 1+i & -1 \\ -i & 1 \end{pmatrix} \hat{\mathbb{R}}$$

Which is shown in Figure 4.

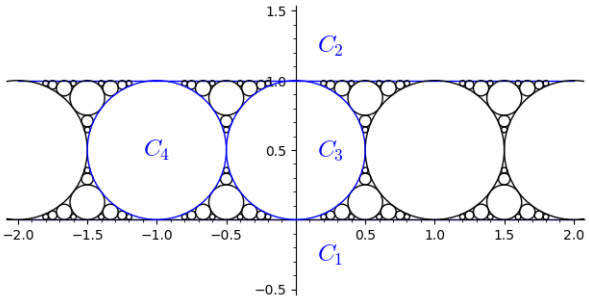


FIGURE 4. The circles C_i

Theorem 3.3. *The map from $\mathbb{P}^1(\mathbb{C})$ to Descartes quadruples up to scaling by $\mathbb{R}^+$ as stated in Lemma 2.4 is the same as taking curvatures of C_1, C_2, C_3, C_4 under the Möbius transformation $z \to \frac{1}{z-(x+yi)}$ or $\begin{pmatrix} 0 & 1 \\ 1 & -(x+yi) \end{pmatrix}$.*

Proof. Say we have the point $z_0 = x + yi \in \mathbb{P}^1(\mathbb{C})$. Let $M = \begin{pmatrix} 0 & 1 \\ 1 & -(x+yi) \end{pmatrix}$ a Möbius transformation. Note that the point z_0 maps to the packing $(y, 1-y, x^2+y^2-y, 1+x^2+y^2+2x-y)$. Using the definition of the four circles in the strip packing C_1, C_2, C_3, C_4 we consider the Möbius transformation on them and what the resulting circle is using Proposition 3.1.

1.) $MC_1 = \begin{pmatrix} 1 & 0 \\ -x-yi & 1 \end{pmatrix} \hat{\mathbb{R}}$ is a circle with curvature $2y$.

2.) $MC_2 = \begin{pmatrix} 0 & 1 \\ 1 & -x+(1-y)i \end{pmatrix} \hat{\mathbb{R}}$ is a circle with curvature $2(1-y)$.

3.) $MC_3 = \begin{pmatrix} -i & 1 \\ (1-y)+xi & -x-yi \end{pmatrix} \hat{\mathbb{R}}$ which is a circle with curvature
$$2\,\mathrm{Im}((1-y-xi)(-x-yi)) = 2(x^2+y^2-y)$$

4.) $MC_4 = \begin{pmatrix} -i & 1 \\ (1-y)+(x+1)i & -1-x-yi \end{pmatrix} \hat{\mathbb{R}}$ which is a circle with curvature
$$2\,\mathrm{Im}(((1-y)-(x+1)i)(-1-x-yi)) = 2(x^2+y^2+2x-y+1)$$

Note that this is just a scalar multiple of the quadruple given by the map from $\mathbb{P}^1(\mathbb{C})$ to ACP up to scaling. Therefore these maps are the same. □

3.2. **Applications.** This alternative map illustrates the geometry that is happening in the parameter space much more easily. This Möbius map takes a point to ∞ and so when we invert the strip packing quadruple through this point, it makes the description in remark 2.5 more obvious. When we invert through a point inside a circle it must produce a bounded packing because this new circle now contains infinity. Similar descriptions can be used for the tangency points, points on the boundary, and limit points.

Holly [Hol21] speaks to this idea of taking a point p to ∞ but determines this method as equivalent to her original question. However, this method can be used to simplify the creation of these fractals by taking this mapping and using a change of coordinates. This method also makes rendering the parameter space much easier because now we can take the strip packing and invert through this point. With the previous map, we only got one quadruple and then had to place these circles in the plane and generate the rest of the packing.

References

[GLM+03] Ronald L. Graham, Jeffrey C. Lagarias, Colin L. Mallows, Allan R. Wilks, and Catherine H. Yan. Apollonian Circle Packings: Number Theory. *J. Number Theory*, 100(1):1–45, 2003. 1, 2

[Hol21] Jan E. Holly. What type of Apollonian circle packing will appear? *Amer. Math. Monthly*, 128(7):611–629, 2021. 1, 3.2

[Koc20] Jerzy Kocik. Apollonian depth and the accidental fractal. https://arxiv.org/abs/2002.04135, 2020. 1

[Ric24] James Rickards. The Apollonian staircase. *International Mathematics Research Notices*, 2024(2):1340–1372, 2024. 1, 2, 2.1, 2.3, 2, 2

[Sta18] Katherine E. Stange. Visualizing the arithmetic of imaginary quadratic fields. *Int. Math. Res. Not. IMRN*, (12):3908–3938, 2018. 3.1

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